

MEGHA CHAKRABORTY

Jirat, Kolkata, West Bengal, India | 9800824732 | megachakraborty2112@gmail.com | LinkedIn: megha-chakraborty-7a6778328 | GitHub: Megha737

PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in data collection, cleaning, and analysis using Python (Pandas), SQL, and SQLite, and working comfort with Linux environments, Git version control, and Docker-based containerization. Built an end-to-end fraud analytics pipeline processing a 6-million-row financial transaction dataset, applying SQL-based anomaly detection logic and visualizing results through an interactive dashboard. Comfortable working with relational databases, REST APIs, exploratory data analysis, and translating raw data into clear, actionable insights. Strong problem-solving foundation from coursework in Data Structures & Algorithms, DBMS, Networking, and Unix/Linux, with the ability to work independently or as part of a team.

EDUCATION

RCC Institute of Information Technology, Kolkata <i>Bachelor of Computer Applications (BCA), 3rd Year — CGPA: 9.05</i> Relevant Coursework: Data Structures & Algorithms, DBMS, OOP using Java, Computer Networks, Operating Systems (Linux/Unix), Software Engineering, Cryptography & Information Security, Advanced Java	Sept 2023 – Present
West Bengal Council of Higher Secondary Education <i>Higher Secondary (XII), Physics / Chemistry / Biology / Mathematics — 70.1%</i>	2022

TECHNICAL SKILLS

- Systems & DevOps:** Linux/Unix, Docker (containerization basics), Git & GitHub, Networking Fundamentals, Bash & Python scripting
- Data Analysis:** Python (Pandas), SQL, Exploratory Data Analysis, Data Cleaning & Preprocessing
- Databases:** SQLite, MongoDB, MySQL, Relational Database Design
- Visualization & Backend:** Recharts, FastAPI, REST APIs
- Programming:** Python, Java, JavaScript, Data Structures & Algorithms
- Web Development:** HTML5, CSS3, React, React Native, Next.js, Node.js, Django, Tailwind CSS

PROJECTS

FinTech Risk & Fraud Analytics Engine (fintech-fraud-analyzer)

Tech Stack: Python (Pandas), SQLite, FastAPI, Next.js, Tailwind CSS, Recharts

- Ingested and structured a 6-million-row financial transaction dataset (Kaggle PaySim) into a relational SQLite database using Python and Pandas.
- Designed and implemented SQL-based anomaly detection logic to flag two fraud patterns: high-frequency low-value transactions ('Salami Slicing') and rapid large-value transfers ('Velocity Spikes').
- Built a REST API with FastAPI and an interactive Next.js/Recharts dashboard to visualize transaction volume, exposure, and flagged anomalies in real time.

LedgerShield — Graph-Based Financial Fraud Detection System

Tech Stack: Python, Django, TigerGraph, HTML5, CSS3, JavaScript

- Built an advanced fraud detection system for the TigerGraph Hackathon that identifies and monitors 'Salami Slicing' fraud, where small unauthorized amounts are systematically shaved off multiple bank accounts.
- Integrated TigerGraph's graph database to analyze complex account and transaction relationships, enabling detection of coordinated multi-account fraud patterns that traditional relational queries miss.
- Developed a live monitoring dashboard (HTML5/CSS3/JavaScript) for real-time visualization of suspicious activity, plus automated reporting that generates high-risk summaries for flagged accounts.

Retail Behavior Engine — Customer Segmentation SaaS Platform

Tech Stack: Python, FastAPI, Pandas, Scikit-Learn, SQLite

- Built an intelligent SaaS platform that applies K-Means clustering to e-commerce transaction data, automatically segmenting customers into behavioral personas (e.g., 'Champions', 'At-Risk') to guide marketing strategy.
- Designed an automated machine learning pipeline (Scikit-Learn) to uncover hidden customer behavioral patterns and a FastAPI backend for real-time processing of uploaded datasets.
- Created a responsive 'Bento-Grid' dashboard for real-time behavioral analytics, with direct CSV export for CRM/marketing tool integration and a guided onboarding tour for new users.

VoiceAI — Multilingual Voice Transcription Platform for Healthcare & Finance

Tech Stack: Python, Next.js, TypeScript

- Built a voice AI platform that transcribes and translates mixed-language speech (code-switched English and regional languages) into standardized English text for healthcare and finance use cases.
- Developed backend modules for audio transcription, processing, knowledge base lookup, and conversation memory (transcriber.py, processor.py, knowledge_base.py, memory.py) to support accurate, context-aware voice understanding.
- Built a Next.js/TypeScript frontend dashboard for reviewing transcribed and translated conversations.

TravelBuddy — Full-Stack Travel Planner

Tech Stack: Next.js, Redux Toolkit, Tailwind CSS, Vite, Node.js

- Developed a full-stack travel planning web application with a responsive UI and state management via Redux Toolkit.

ACHIEVEMENTS

Hackathons & Competitions

- Selected for the Smart India Hackathon (SIH), engineering "DisasterReady," a digital platform for disaster management education and virtual drills.
- Developed "LedgerShield" at the TigerGraph Hackathon, a graph-based fraud detection solution identifying 'Salami Slicing' transactions.
- Built "Retail Behavior Engine," a customer segmentation SaaS platform using K-Means clustering for e-commerce behavioral analytics.
- Team Lead of "Team Kill Bill" — applied to the HBTM Hackathon (IABTM x IIIT Pune, Agentic AI for Human Potential) and HackIndia Spark 6.

INTERNSHIP EXPERIENCE

Web Developer Intern — Indian Institute of Internship

Jul 2025 – Aug 2025

Project-based industrial training | Technology: Django

- Built and deployed a full-featured travel website, handling both frontend and backend development along with database integration.
- Certificate ID: RCC020, issued 10th September 2025.

CERTIFICATIONS

- Software Programmer – Python, Nasscom FutureSkills Prime (IT-ITeS SSC) — March 2026
- Certificate of Completion – Django, Indian Institute of Internship — September 2025
- Introduction to Generative AI – Google Cloud, via Simplilearn — April 2025
- Certificate of Participation – India AI Impact Festival 2025 (Ministry of Education, NITI Aayog, NIELIT, Intel) — Nov 2025